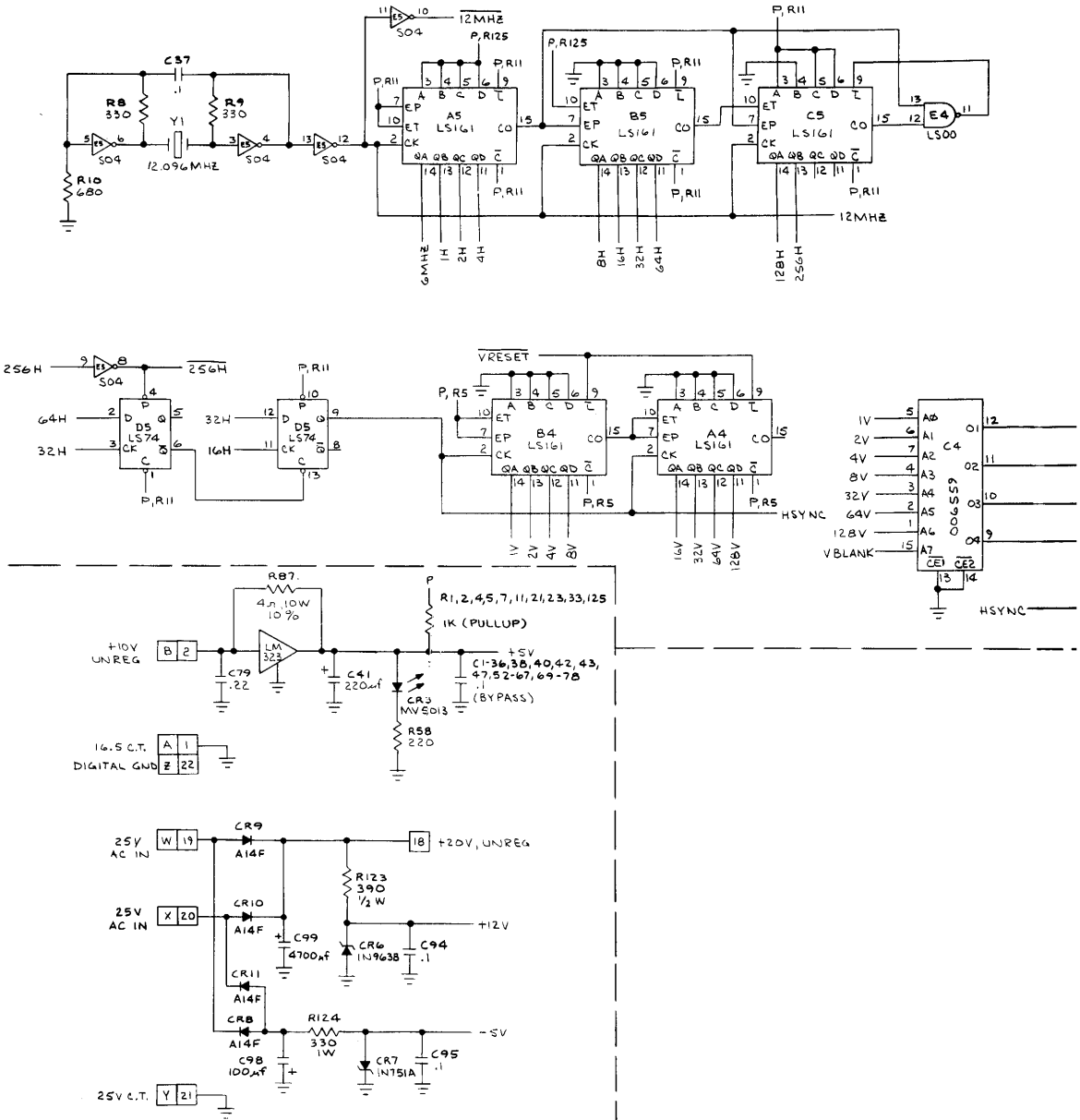
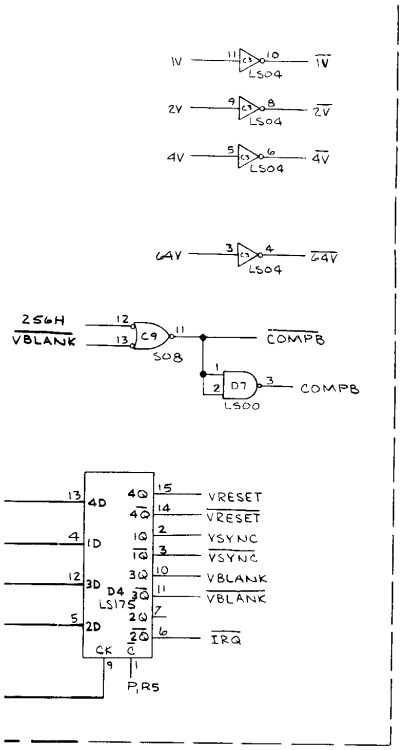
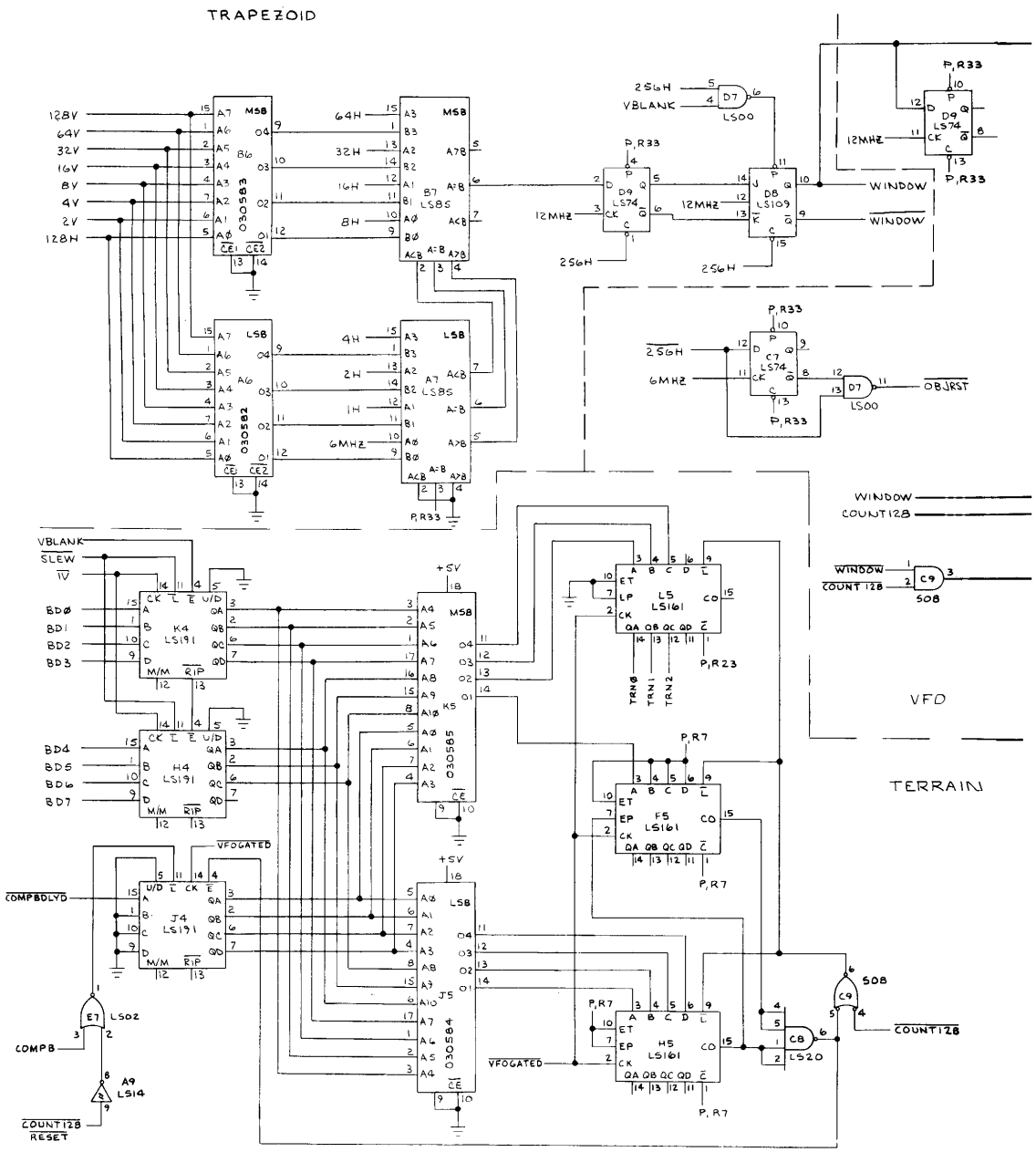

C37
.1
R8
330
R9
330
Y1
12.096MHZ
R10
680
S04
12MHZ
P,R125
P,R11
A5
LS161
B5
LS161
C5
LS161
LS00
6MHZ
1H
2H
4H
8H
16H
32H
64H
128H
256H
256H
64H
32H
D5
LS74
32H
16H
D5
LS74
VRESET
P,R5
B4
LS161
A4
LS161
HSYNC
1V
2V
4V
8V
16V
32V
64V
128V
VBLANK
C4
006559
HSYNC
R87.
4.7Ω 10W
10%
R1,2,4,5,7,11,21,23,33,125
1K (PULLUP)
+10V
UNREG
LM
323
C79
.22
C41
220mf
+5V
C1-36,38,40-42,43,
47,52-67,69-78
.1
(BYPASS)
CR3
MV5013
R58
220
16.5 C.T.
DIGITAL GND
25V
AC IN
CR9
A14F
+20V, UNREG
R123
390
½ W
25V
AC IN
CR10
A14F
+12V
C99
4700mf
CR6
IN9638
C94
.1
CR11
A14F
CR8
A14F
R124
330
1W
-5V
C98
100mf
CR7
IN751A
C95
.1
25V C.T.

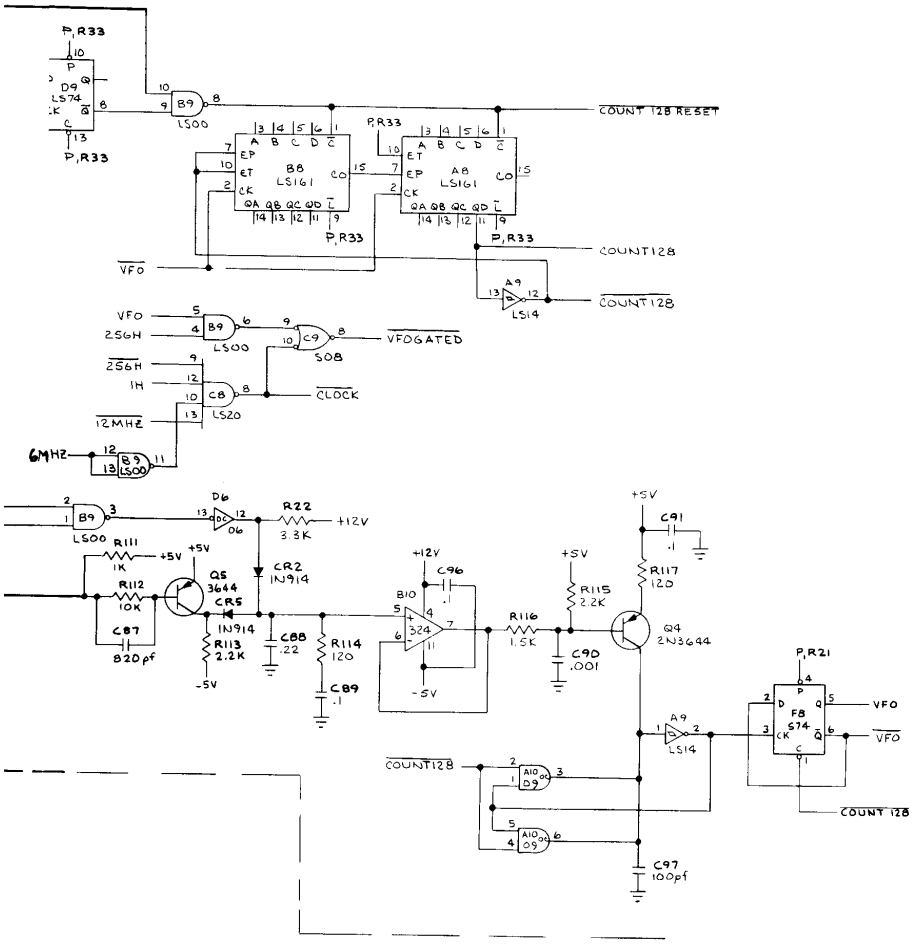


SYNC, POWER

LAST DES	NOT USED
C99	C39
R127	
CR11	
Q6	
Y1	
S3	

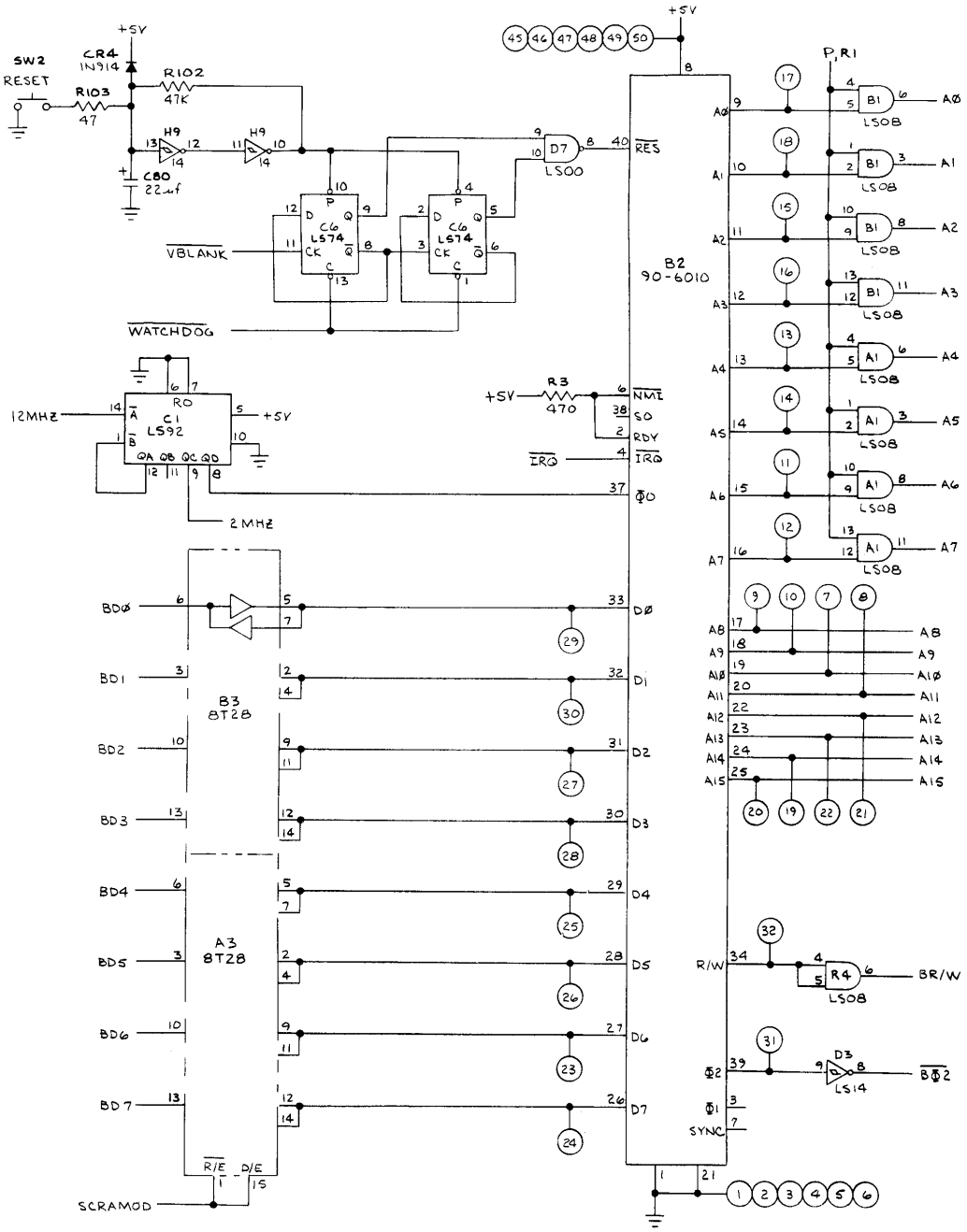
Figure 4-2 Sky Raider PCB Schematic Diagram Sheet 1 of 7


TRAPEZOID
128V
64V
32V
16V
8V
4V
2V
128H
030583
B6
B7
LSB5
030582
A6
A7
LSB5
64H
32H
16H
8H
4H
2H
1H
6MHZ
P,R33
256H
VBLANK
D7
LS00
12MHZ
D9
LS74
WINDOW
D8
LS109
C7
OBJRST
WINDOW
COUNT128
C9
S08
VBLANK
SLEW
1V
BD0
BD1
BD2
BD3
BD4
BD5
BD6
BD7
K4
LS191
H4
J4
K5
030585
J5
030584
+5V
VFOGATED
COMPBDLVD
COMPB
E7
LS02
A9
LS14
COUNT128
RESET
L5
LS161
P,R23
TRN0
TRN1
TRN2
VFD
P,R7
F5
H5
TERRAIN
C8
LS20



TERRAIN, VFO, TRAPEZOID

Figure 4-2 Sky Raider PCB Schematic Diagram Sheet 2 of 7


SW2
RESET
CR4
IN914
R102
47K
R103
47
C80
22uf
+5V
H9
VBLANK
WATCHDOG
C6
LS74
D7
LS00
RES
B2
90-6010
P,R1
B1
LS08
A1
LS08
A0
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
A11
A12
A13
A14
A15
C1
LS92
12MHZ
2MHZ
R3
470
NMI
SO
RDY
IRQ
Φ0
B3
8T28
A3
8T28
BD0
BD1
BD2
BD3
BD4
BD5
BD6
BD7
D0
D1
D2
D3
D4
D5
D6
D7
R/E
D/E
SCRAMOD
R/W
R4
BR/W
Φ2
D3
LS14
BΦ2
Φ1
SYNC

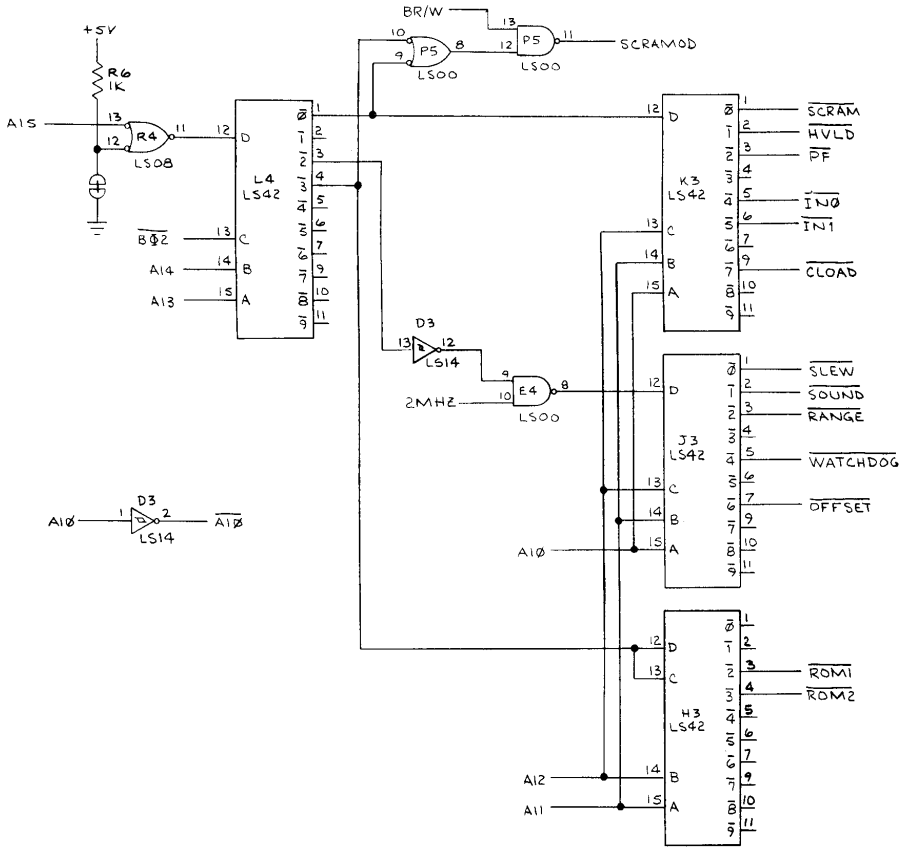
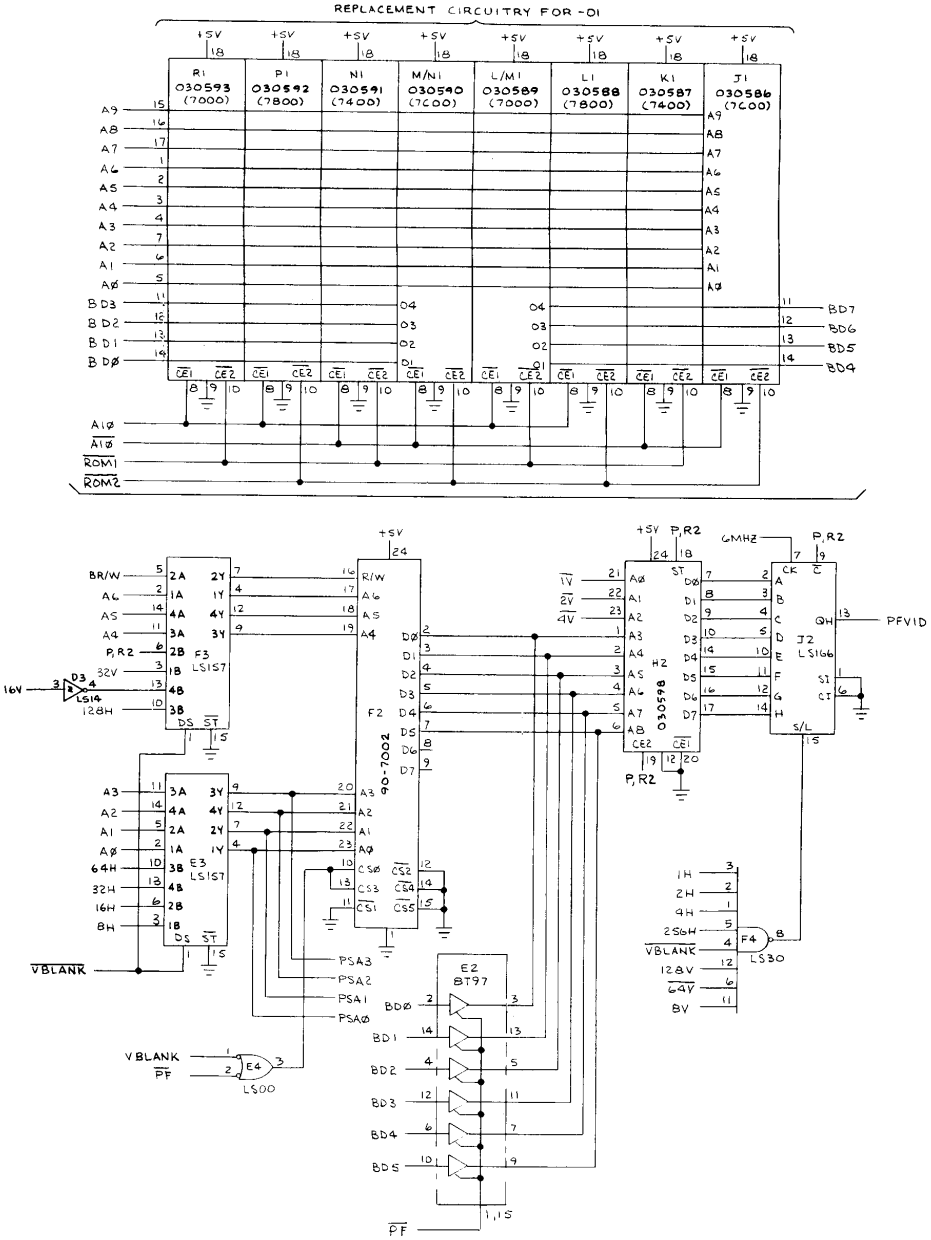


Figure 4-2 Sky Raider PCB Schematic Diagram
Sheet 3 of 7


REPLACEMENT CIRCUITRY FOR -01
R1 030593 (7000)
P1 030592 (7800)
N1 030591 (7400)
M/N1 030590 (7600)
L/M1 030589 (7000)
L1 030588 (7800)
K1 030587 (7400)
J1 030586 (7600)
+5V
A9
A8
A7
A6
A5
A4
A3
A2
A1
A0
BD3
BD2
BD1
BD0
BD7
BD6
BD5
BD4
A10
A10
ROM1
ROM2
BR/W
P,R2
32V
16V
128H
F3 LS157
LS14
E3 LS157
64H
32H
16H
8H
VBLANK
F2
90-7002
E2 8T97
PSA3
PSA2
PSA1
PSA0
VBLANK
PF
E4 LS00
1V
2V
4V
H2
030598
6MHZ
J2 LS166
PFVID
S/L
1H
2H
4H
256H
128V
64V
8V
F4 LS30

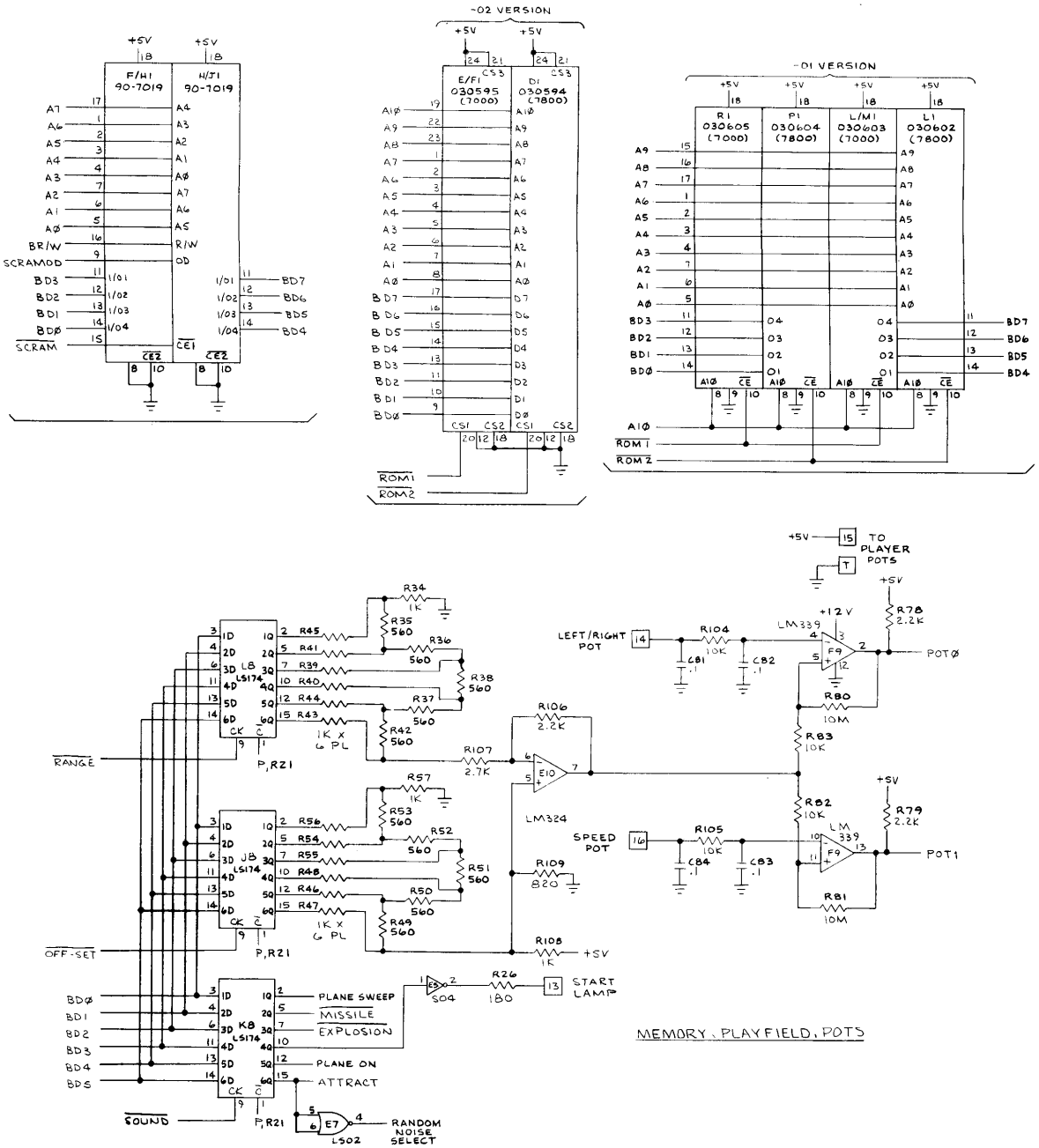
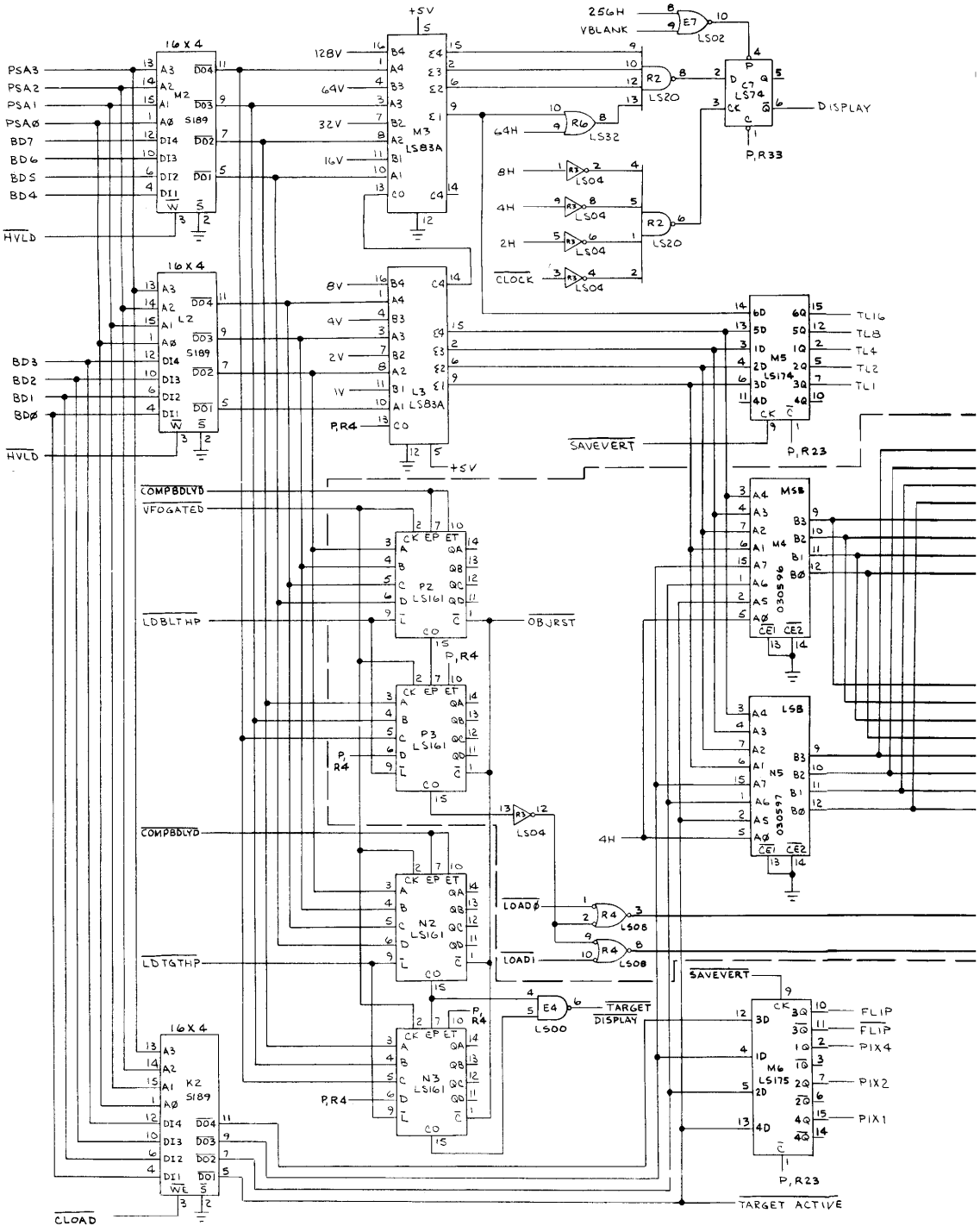
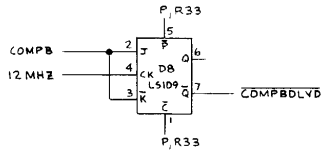
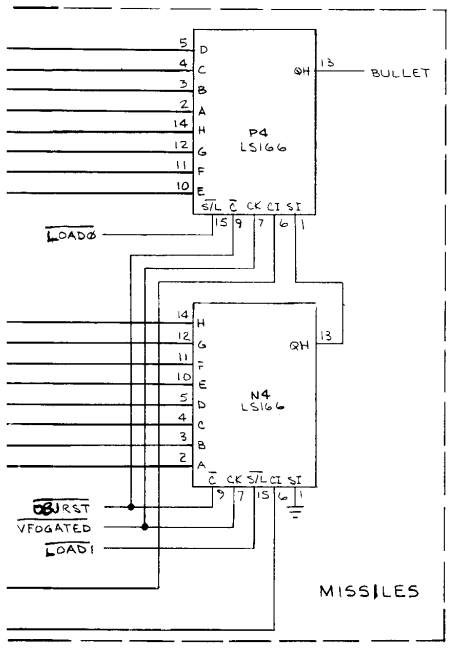


Figure 4-2 Sky Raider PCB Schematic Diagram Sheet 4 of 7


PSA3
PSA2
PSA1
PSA0
BD7
BD6
BD5
BD4
HVLD
16 X 4
M2
S189
BD3
BD2
BD1
BD0
L2
128V
64V
32V
16V
+5V
M3
LS83A
256H
VBLANK
E7
LS02
R2
LS20
R6
LS32
64H
C7
LS74
DISPLAY
P,R33
8H
4H
2H
CLOCK
LS04
8V
4V
2V
1V
L3
P,R4
TL16
TL8
TL4
TL2
TL1
M5
LS174
SAVEVERT
P,R23
COMPBDLYD
VFOGATED
P2
LS161
LDBLTHP
OBJRST
P3
N2
N3
LOAD0
LOAD1
LDTGTHP
E4
LS00
TARGET DISPLAY
M4
M5B
LSB
N5
M6
LS175
FLIP
PIX4
PIX2
PIX1
TARGET ACTIVE
K2
CLOAD

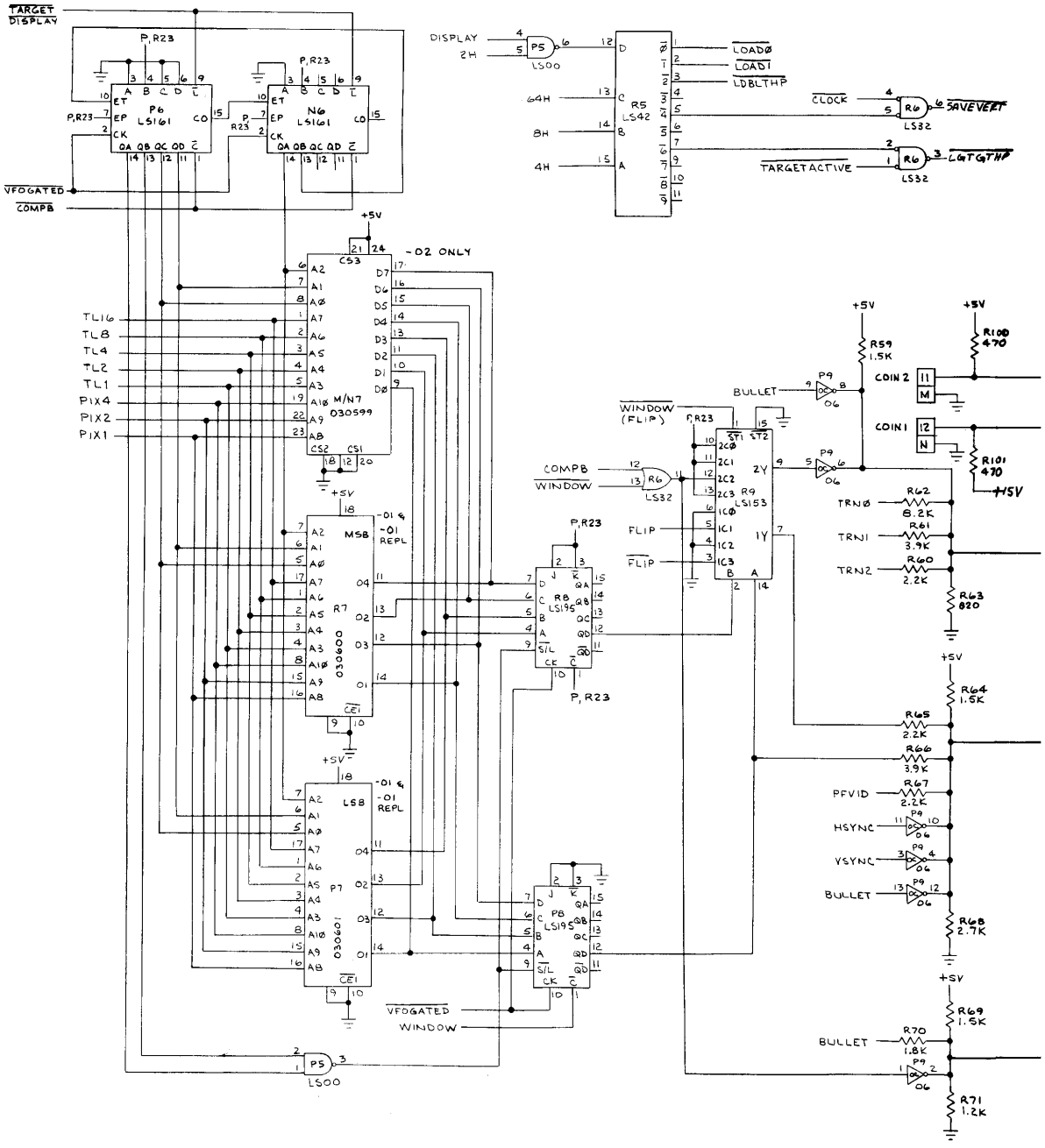


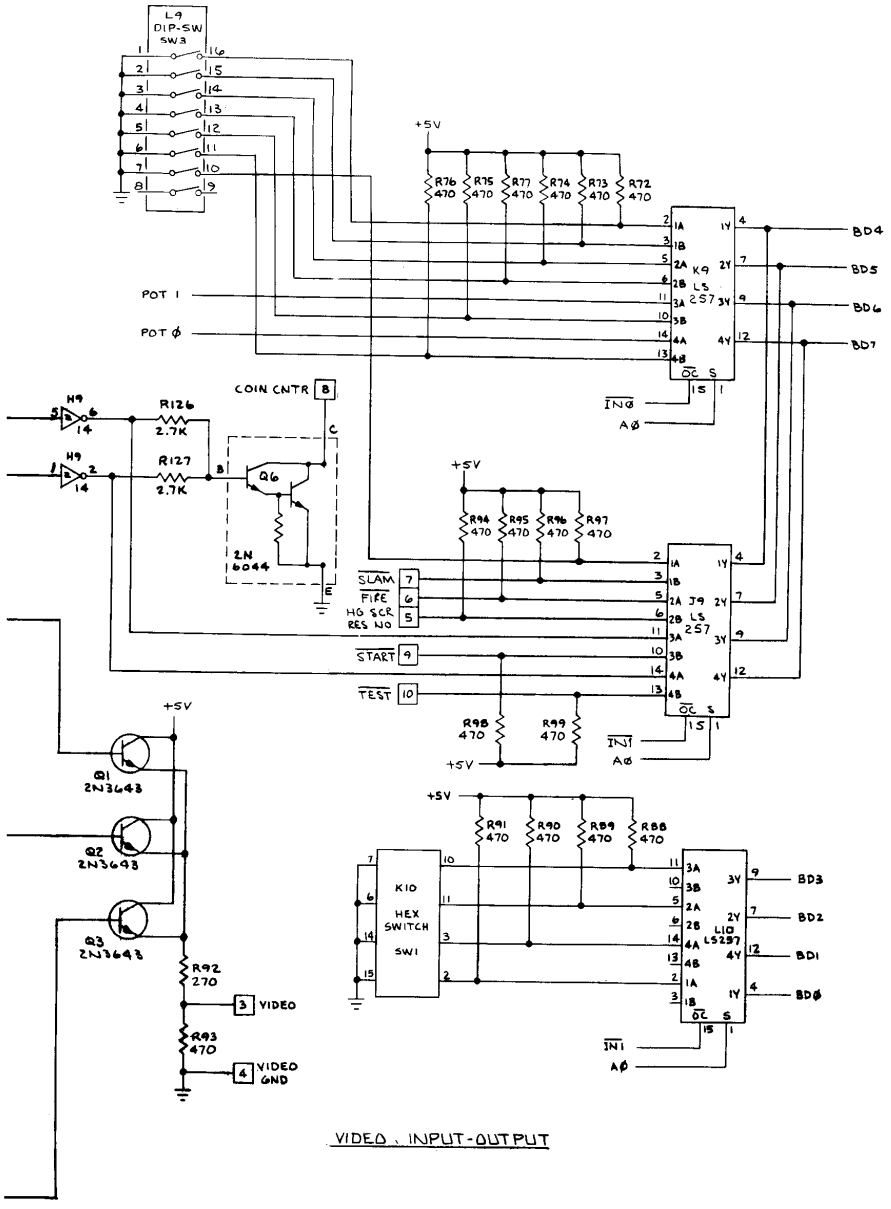


MOVING
OBJECT
CONTROL

MOVING OBJECT CONTROL, MISSILES

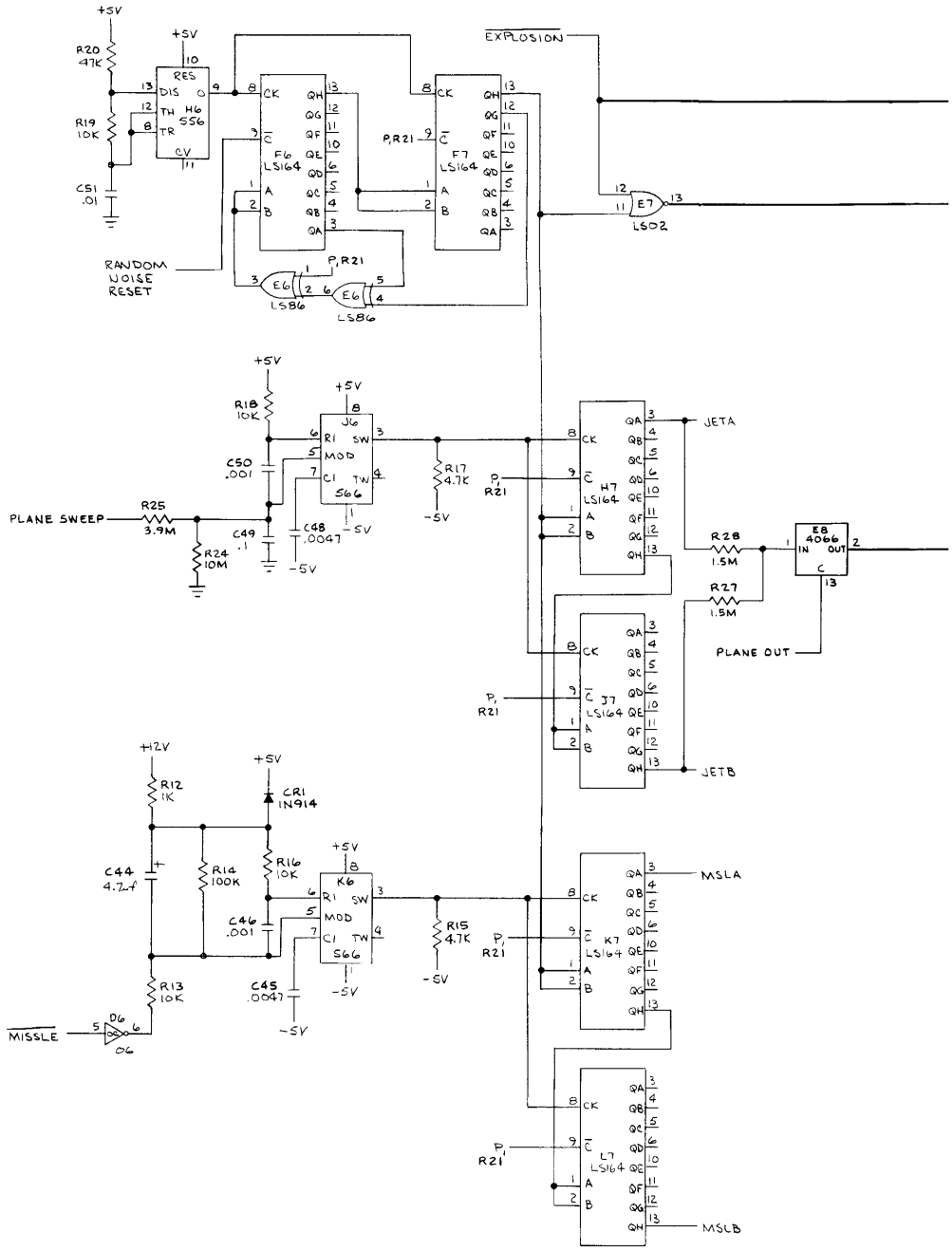
Figure 4-2 Sky Raider PCB Schematic Diagram
Sheet 5 of 7

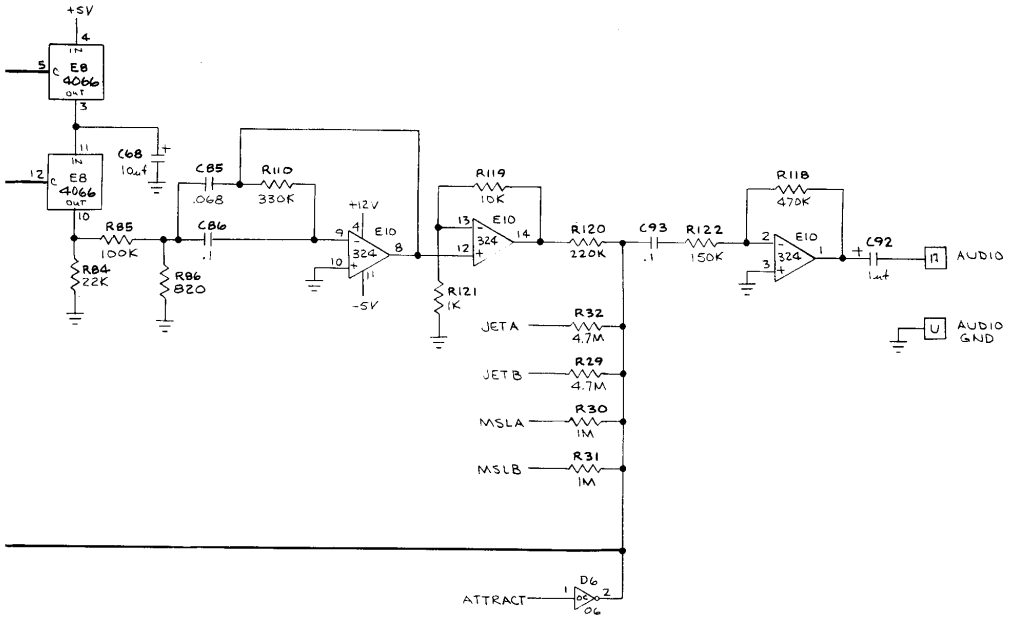

TARGET DISPLAY
P,R23
P6 LS161
N6 LS161
VFOGATED
COMPB
DISPLAY
2H
P5
LS00
64H
8H
4H
R5 LS42
LOAD0
LOAD1
LDBLTHP
CLOCK
R6
LS32
SAVEVERT
TARGETACTIVE
LOADTHP
+5V
CS3
-02 ONLY
TL16
TL8
TL4
TL2
TL1
PIX4
PIX2
PIX1
M/N7 030599
MSB
-01 & -01 REPL
R7 030600-01
LSB
P7 030601-01
P8 LS195
WINDOW
VFOGATED
WINDOW (FLIP)
BULLET
COMPB
WINDOW
FLIP
R9 LS153
R59 1.5K
R100 470
COIN2
COIN1
R101 470
TRN0
TRN1
TRN2
R62 8.2K
R61 3.9K
R60 2.2K
R63 820
R64 1.5K
R65 2.2K
R66 3.9K
R67 2.2K
PFVID
HSYNC
VSYNC
R68 2.7K
R69 1.5K
R70 1.8K
R71 1.2K



VIDEO . INPUT-OUTPUT

Figure 4-2 Sky Raider PCB Schematic Diagram
Sheet 6 of 7


+5V
R20 47K
R19 10K
C51 .01
H6 556
RES
DIS
TH
TR
CV
F6 LS164
F7 LS164
P,R21
EXPLOSION
RANDOM NOISE RESET
E6 LS86
E7 LS02
R18 10K
C50 .001
J6 566
MOD
R17 4.7K
-5V
PLANE SWEEP
R25 3.9M
R24 10M
C49 .1
C48 .0047
H7 LS164
JETA
R28 1.5M
E8 4066
R27 1.5M
PLANE OUT
J7 LS164
JETB
+12V
R12 1K
CR1 IN914
C44 4.7uf
R14 100K
R16 10K
C46 .001
K6 566
R15 4.7K
C45 .0047
R13 10K
MISSLE
D6
K7 LS164
MSLA
L7 LS164
MSLB



AUDIO

Figure 4-2 Sky Raider PCB Schematic Diagram Sheet 7 of 7